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Design Optimization and Virtual Prototyping of Hybrid Solar Car Systems

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***Abstract* - This research presents a parametric investigation and simulation-based evaluation of advanced hybrid solar car architectures as a sustainable alternative to conventional vehicles. A detailed CAD model was developed in SolidWorks, and system-level performance was analyzed using ANSYS14 for load, stress, and strain conditions for both frontal and rear impacts. Parametric results for energy flows, load paths, and frame resilience are discussed. The study finds that, while solar-hybrid technology offers distinct benefits such as reduced emissions and fuel consumption, technical and economic deployment barriers still persist. Nevertheless, with careful mechanical and electronic integration, such vehicles demonstrate clear potential for future widespread adoption over fossil-fuel platforms.**

***Keywords* - Hybrid solar vehicle, solar energy, IC engine, Electric operated system**

I. Introduction

The brief study of hybrid solar car is efficient in our daily life because now day's pollution and fuel rate is very big problem many people having fuel cars. Use of solar energy is being used for car, besides the control of vehicular pollution in the city, less consumption of fuel, Hybrid solar car are effective reducing global warming and environment problem in big frame. In the present work, the objective of this work is to estimate the potential of both energy as PV energy and mechanical engine power, both powers will be utilized in running car with weight reduction can be achieved primarily by the introduction of better material, design optimization and better manufacturing processes. The hybrid solar car is one of the potential items for weight reduction in hybrid solar car as it accounts for 5% - 10% of the weight. Various advantages for hybrid solar car by using solar technology

1. Reduction in conventional car demand in urban city
2. Minimum the pollution problem in urban city
3. Give clean energy which will reduce the carbon dioxide emission every month
4. Reduction in fuel demand

A hybrid car is a vehicle which can be used three power sources are a solar energy with electric motor, electrical operated and a small combustion engine to run a car.

They are slowly gaining popularity with the auto buyers because they are seeing the benefits that owning a car will help them to reduce their carbon emission and is also energy efficient. It is also healthful for environment. In this paper, we are firstly creating a drawing and design of hybrid solar car by using CATIA V5R19 tool after that we performed using the finite element method(FEM) done using ANSYS 13.0software. Modeling was done for front and rear unit frame 3D brick element (solid 45) and fivenode.

II. Literature Review

In this paper present with a detailed study of optimal sizing, fuel consumption of a solar car based on a longitudinal vehicle dynamics mode and energy flow, weight, overall cost of vehicle. It is shown that fuel saving can be achieved for intermittent use with average power and economical feasibility [1][3]. Hybrid Solar Vehicles (HSV), derived by integration of Hybrid Electric Vehicles with Photo-Voltaic sources, may represent a valuable solution to face both energy saving and environmental issues, particularly in urban driving. [6][8]. This paper is also focuses on the general, technological issues and challenges ahead of plug-in hybrid electric vehicles in relation to major components

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which can be used for detail of design consideration and selection of component for electric motor and battery bank, control strategy. Other technical challenge as light weight material used in a vehicle, low resistance tire and better aerodynamic structure of its .it type of vehicle is importance of economies and successful deployment of this plug- in hybrid technology [8] we investigate the use of photovoltaic systems as auxiliary power generators in hybrid and electric vehicles. This technology provides an as yet unexploited possibility with the advantages of a new power source, which is light, noiseless, maintenance-free and continuously working. A notable reduction of air emissions can be achieved through a synergy of various technological breakthroughs, such as the method we present of introducing photovoltaic arrays and additional electrochemical energy storage capacity in vehicles. Solar cars are also considered as a case study in order to demonstrate the use of solar panels in electric cars [15].

A comparative study of three different configurations for a hybrid solar vehicle has been recently presented and development at the University of Salerno in 2008[16]. The hybrid electric vehicle (HEV) system that consists of mechanical engine, electric motor and batteries has gained increased interests because of the interest's drawback of IC engine such as pollution. For used Advance control strategy of control the pollution level of city by using total electric operated system of vehicle and perform the preliminary test and road test of a vehicle. [20]

III. Related Theory

A. Hybrid solar car Components -Electric vehicle was first designing and developing by the Baker Motor Company since 1990s. A main advantage of EV over the internal combustion engine can thus be exploited in terms of no carbon emissions occurred due to only use the electric motor to drive the engine [10]. Generally, the electric car and hybrid solar car is consisted of six main parts: PV panel, electric motor, electric Generator, battery bank, IC engine and electronic controlling unit (ECU) respectively. A diagram of the hybrid solar car can thus be illustrated in Fig. 1.

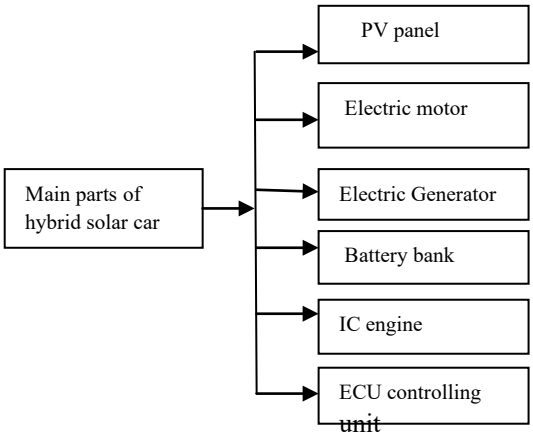

Figure 1 Main components of hybrid solar car

(1) PV panel convert sunlight into electricity, which is stored in batteries. Then its energy will be utilized for hybrid car propulsion.

(2) Electric motor has, generally, been employed for driving the hybrid solar car (HSC). However, we can observe that the brushless DC motor (BLDC) is often operated in the (HSC)over the classical DC motor due to long lifetime operation, high speed and also high torque.

(3) Electric Generator is self energy generated in a car during a regeneration brake or slop on a road.

(4) Battery bank is an important component for the HSC. It has been generated 24 V DC for supplying to the electric motor and also electronic devices in the HSC.

(5) IC engine has been employed for driving of hybrid solar car which can be used during a night or minimum electric energy of HSC.

(6)Electronic controlling unit (ECU) is an electronics circuit that is used for controlling the energy in the electric motor which can be provides a speed variation.

B. Transmission system – In this hybrid solar car, we are using two types of transmission system as below with a diagram of the transmission system of hybrid solar car can thus be illustrated in Fig. 2.

Front wheels

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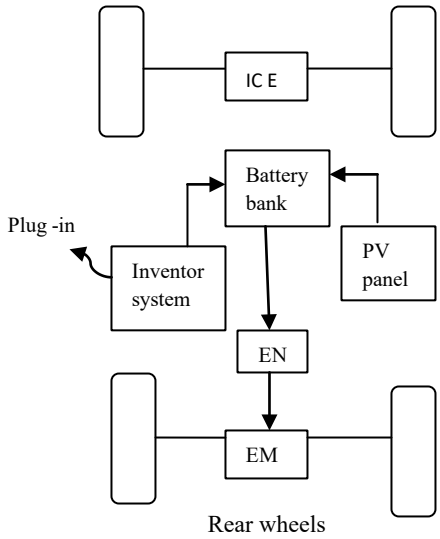

Figure 2. A general Transmission system of hybrid solar car

Where EM= electric motor, EN= electric control node which can be control the supplies of voltage, PV panel, ICE= internal combustion engine (petrol operated 4-stroke engine), Battery bank, plug –in with inventor system.

Two type of transmission system

1. Front drive system –The IC engine is placed on the front of the car. With the help of chain drive mechanism, the power stored in crank shaft is driven to the axle of front wheels.
2. Rear drive system – The electric motor is placed near the rear axle of hybrid solar car which can be operated by charged batteries then electric motor rotate the rear axle of rear wheels

Both systems are placed in hybrid solar car, at a onetime one system can be used in transmission of power to wheels as either rear or front wheels of a hybrid solar car. If one system gets fail then we have an option to choose from alternative driving mechanism.

IV. Model And Design

Hybrid solar car is the location of solar panel across on roof, bonnet and boot section of a car, at almost horizontal and vertical potion of a car. In a general model, it could be consider of at least two additional options as

1. Horizontal panel
2. Vertical panel

It is designed by using in CATIA V5R19 software. Firstly, we design a framework and then we are design different parts of solar car like as solar panel, IC engine, transmission system, braking system, wheels and axles, steering system and dashboard. All these system is design after it will be assembled in frame of car. The drafting and design of hybrid solar car is also shown in figure 3 (a) and figure 3 (b).

Figure.3 (a) Drafting and drawing of hybrid solar car

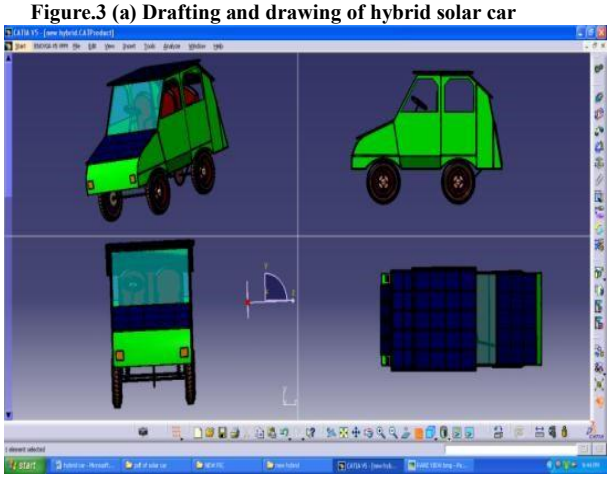

Figure 3 (b) Different view of hybrid solar car with the help of CATIA V5R19 software.

A. Design of PV system

The PV system design is based on the size and capacity of load. The energy produced by PV panel obtained from two types of energy contribution as shown in equation 1 and equation 2 form by the adopted in G.Rizzo and Ivan Arise paper as[6][7]

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1. *Driving time* - the energy produced during a running period of a car is calculated by the formula

$$E_d = \eta_p \, A_p \, e_{sun} \, \frac{h_{sun} - h_d}{h_{sun}} \, \alpha$$

2. *Parking time* - The energy produced during the parking period of a car is calculated by the formula

$$E_p = \eta_p \, A_p \, e_{sun} \, \frac{h_{sun} - h_d}{h_{sun}} \, \beta$$

Where η_p = PV panel efficiency, A_{pv} = PV surface area, e_{sun} = the average energy daily enough by solar panel captured (4.3 K hr/day), h_{sun} = solar energy captured during a sun at (7 AM to 6 PM), h_d =solar energy during a parking time

Total power = Driving time + Parking time

$$E = Ep + Ed$$

B. Design of Battery system

In the design we will take depth of discharge to be 75%. temperature correction is needed because at low temperature battery efficiency decreases.

Battery bank capacity in Ampere hour (AH) is given by

$$B_{rc} = E_{c\,(Ah)} \times D_s / (DOD)_{max} \times \acute{\eta}$$

Where, DOD =Battery depth of discharge

D_s = battery autonomy or storage days $\acute{\eta}$ = temparetrur correction factor = 0.9 $E_{c\,(Ah)}$ = energy or load is given by Ampere in hour Batteries in parallel is given by

$$B\,p = Brc / Bsc$$

Where, Bsc = capacity of selection battery (Ah)

Batteries in series is given by

$$Bs = Vn/Vs$$

Where,Vn = nominal voltage of battery

Vs = nominal voltage of system

Total battey bank of system

$$BT = Bp \times Bs$$

C. Simulation and Analysis of frame or body of Hybrid solar car

To design hybrid solar car frame, a stress analysis was performed using the finite element method done using ANSYS software. Modeling was done for front and rear unit frame 3D brick element (solid 45) and five-node. Also, analysis carried out for hybrid solar car frame which can bear maximum load and shear stresses along the each section of car measured). The maximum load and shear stresses along the bonded adhesive layer for glass/epoxy were measured different collision as shown in figure 4 to figure 8 in below

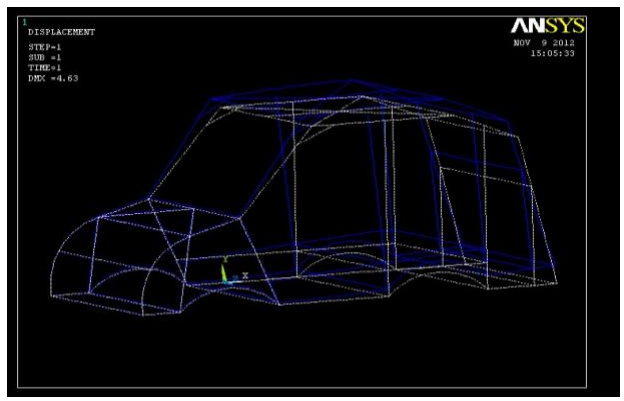

Figure 4. Displacement of rear collision

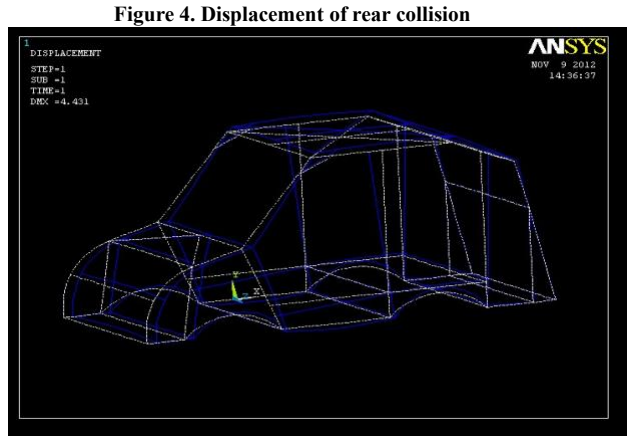

Figure 5. Front collision of car frame collision

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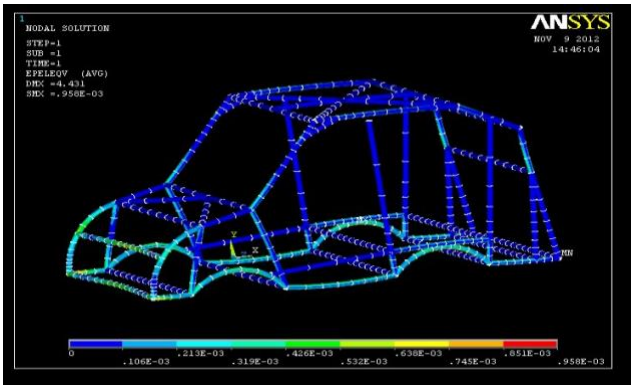

Figure 6. Front collision of car elastic strain, von- mises

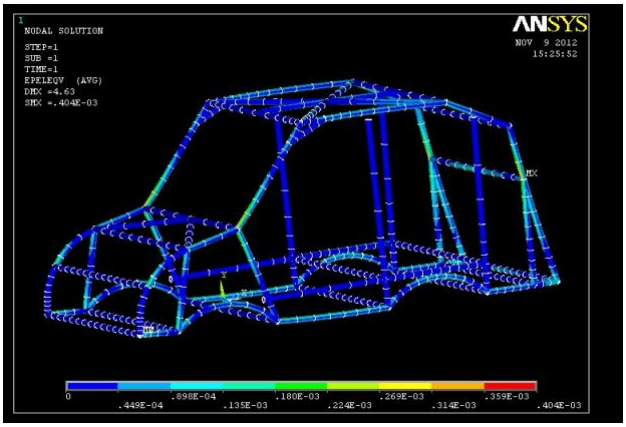

Figure 7. Rear collision of car elastic strain, von- mises

Last figure 8 is given a detail of front and rear collision of hybrid solar car elastic stress, von-mises.

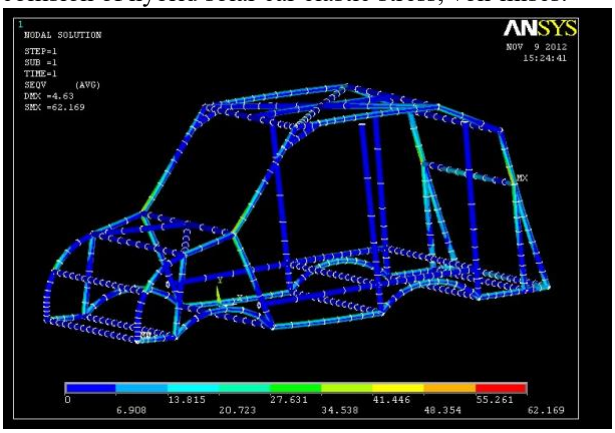

Figure 8. Rear collision of car elastic strain, von- mises

V. Results And Discusion

A. General Dimension of hybrid solar car

We find out the general dimensions of a hybrid solar car by the help of drawing and drafting software using in a CATIA V5R19. A table of the hybrid solar car can thus be illustrated in table 1

Table 1
Showing a general dimension of a hybrid solar car

S. No.	Dimensions of hybrid solar car	Length in mm
1	Wheel Base	2813 mm
2	Wheel track	1160 mm
3	Height of car	1221.46 mm
4	Ground clearance	280 mm

B. Area and Total power generated by PV panel

Total Area of PV panel = 835396 (mm^2) = 0.835 $meter^2$

No of panel mounted in roof = 3, single-single panel mounted in bonnet and backside of car

Total power generated by panel = No. PV panel X Power

= 5 X 150Watt
= 750 Watt
= 0.75 K Watt

C. General batteries required in system

All the information about the battery system and total load is required by electric motor is given in table 2.

And battery is general used in an auto- vehicle for different purpose on present day.

Used in a hybrid solar car battery system can be design a general requirement of car running (load).

Table 2
Showing a general batteries required in system

Capacity of Required by load	2 Kw
Capacity of battery (Ah)	5 Ah
Batteries in parallel	1
Batteries in series	4
Total No. of battery	4
Power of batteries	220- 240 watt

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D. Analysis of frame and structure of hybrid solar car was performed using the finite element method done using ANSYS software. Modeling was done for front and rear unit frame 3D brick element (solid 45) and five-node. And data is given in table

Table 3
Showing analysis results of frame /body of hybrid solar car

Types of collision of hybrid solar car	Max. displacement (mm) about the test axis	Von –Mises Elastic strain	Von-Mises stress (MPa)
Front collision	4.431 mm	0.958E-03	47.32
Rear collision	4.630 mm	0.404E-03	62.169

And graphical shows a comparison between conventional cars over hybrid solar car. It is comparison in two basic as shown in below

1. Fuel consumption based
2. Emission level based

Both figure 9 and figure 10 shows the HSC are the best solution which solve the major problem of fuel consumption and pollution level in urban city .This type of car minimize all problem of conventional

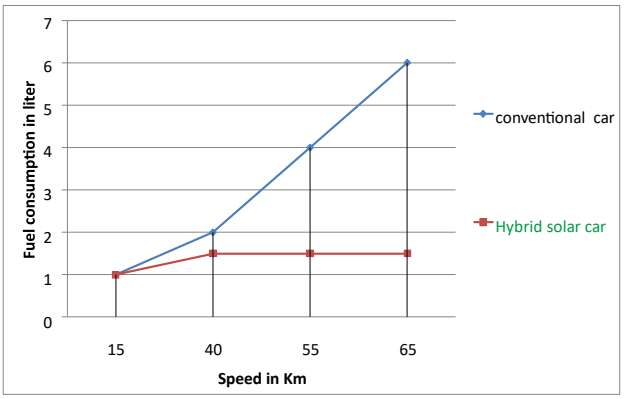

Figure 9.Shows Comparison between the fuel consumption in conventional car over hybrid solar car

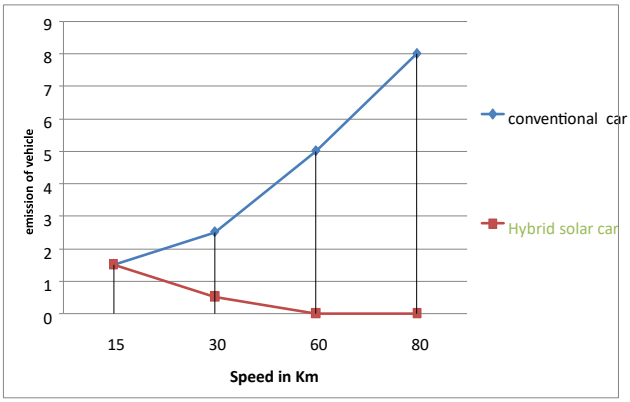

Figure 10. Shows Comparison between the emission level in a conventional car and hybrid solar car

VI. Conclusion

Suitable design based analysis of hybrid solar car has been given and results of battery bank and sizing, total area of car which can be used in PV array, capacity of total load and analysis of car body have been tabulated.

It will be used for research work and education purpose. In future this type of car have lot of marketing value because it will be used non renewable resource and renewable energy .This type of car does not create any pollution so it is also have a lot of positive point toward nature . Only the manufacturing cost is high but maintenance cost is almost zero. Hence this car is economical and environmental friendly

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